

1 -Year Diabetes Progression Risk

The Bottom Line

Evaluating specific physical measurements alongside blood lipid chemistry allows health services to accurately identify nearly 50% of the explainable risk for 1-year diabetes progression.

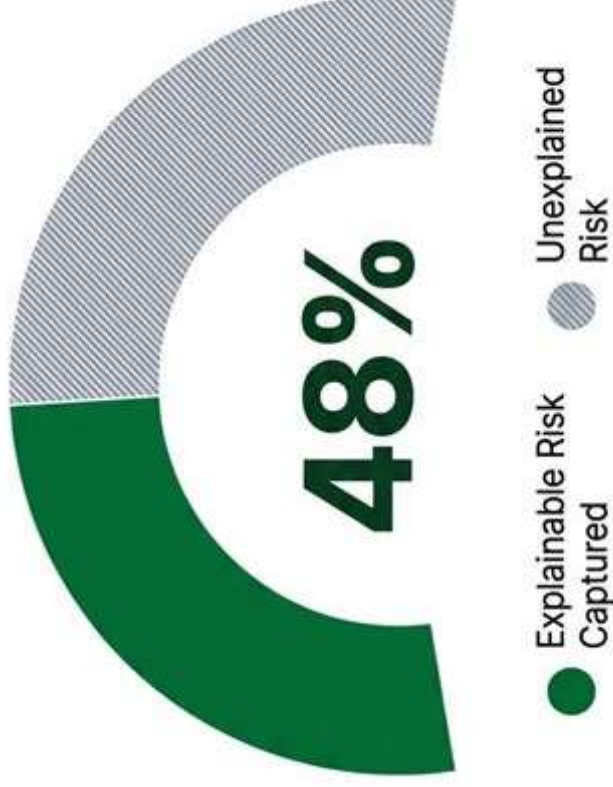
The Data

A dataset consisting of 442 patient observations collected at baseline,

Improvement

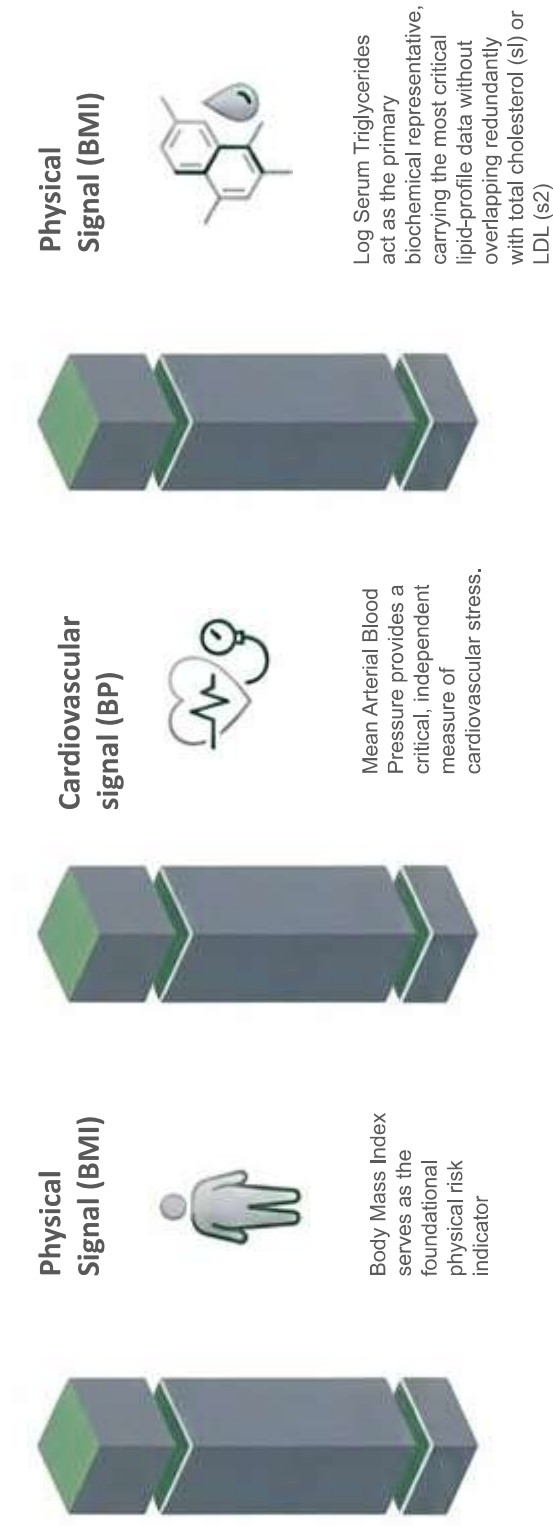
Relying on body weight (BMI) alone for population risk screening is a flawed, incomplete strategy. A targeted composite approach unlocks meaningful population- level insight for preventive resource allocation.

Gauge of Explainable Disease Risk



Three Core Signals Drive the Predictive Framework

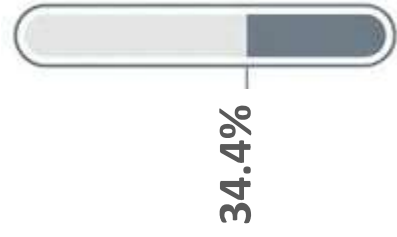
Out of 10 baseline metrics evaluated (including age, sex, and multiple serum measurements), a specific triad of indicators provided the strongest, most stable predictive value for disease progression.



The Advantage of a Composite Clinical Score

Single-Metric Screening

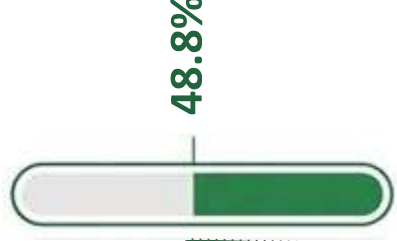
BMI Alone



A simple regression model artificially inflates the apparent effect of BMI because it absorbs the shared, hidden variances actually caused by other factors

Multiple-Metric Approach

BMI + BP + S5



By evaluating these metrics together, we control for overlapping variances. We isolate the true marginal effect of each factor, increasing overall accuracy by nearly 14%

14% Improvement

Critical Evidence Boundaries and Clinical Red Lines



No Direct Causality

Medical interventions that artificially alter BMI or BP cannot be assumed to yield direct linear reductions in progression



Unmeasured Confounders

Crucial clinical determinants were unmeasured and may introduce confounding bias.



Model Uncertainty

Only 48% of the change in progress was explained by these three factors

Improving Model Reliability

